EXPERIMENT 3: CISCO PACKET TRACER

AIM

a) To understand the environment of Cisco Packet Tracer and design a simple network.
b) To analyse the behaviour of network devices using the Cisco Packet Tracer simulator.

REQUIREMENTS

Cisco Packet Tracer, 4–6 PCs, a hub, a switch, and copper straight-through cables.

PROCEDURE

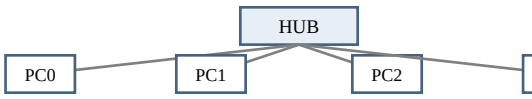
1. Place the required PCs and network device in the Logical workspace. Connect the PCs to the device using copper straight-through cables.
2. Configure each PC with a unique IP address in the same subnet.
3. Build separate topologies using only a hub, only a switch, and both devices together.
4. Use Add Simple PDU to send a packet between two PCs. In Simulation mode, observe packet movement and device responses.

IP CONFIGURATION

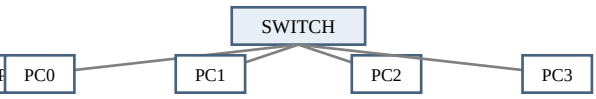
Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

NETWORK TOPOLOGIES

1. Hub-only network



2. Switch-only network



3. Combined hub-and-switch network



OBSERVATIONS

Hub: The hub repeats incoming data to all connected ports. Therefore, all connected hosts can receive the transmitted frame, and unnecessary traffic is generated.

Switch: The switch learns MAC addresses and forwards frames to the appropriate destination port when the destination is known. This reduces unnecessary traffic compared with a hub.

Simulation: When a PDU is sent from one PC to another in the same subnet, Packet Tracer shows the packet travelling through the connected device. A successful PDU indicates that the configured hosts can communicate.

College network topology: A typical campus network uses access switches to connect lab computers and other end devices, with uplinks to distribution/core switches and routers for communication between networks. The exact arrangement depends on the college network design.

STUDENT OBSERVATION

a) Forwarding behaviour of hub and switch: A hub broadcasts incoming data to all connected ports, whereas a switch forwards frames to the destination port using its MAC address table.

b) College network topology: A typical college LAN follows a hierarchical star topology, where computers connect to access switches that uplink to core switches and routers. The exact topology depends on the college's actual network layout.

RESULT

The Cisco Packet Tracer environment was studied, and hub-only, switch-only, and combined networks were designed and simulated. The forwarding behaviour of the hub and switch was observed and compared.

PACKET TRACER SCREENSHOT

A simple point-to-point network created in Cisco Packet Tracer.

